

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with minimal payment edges. Fail: Sum of debts does not equal sum of credits.	This is the core feature of the app — it reduces the number of repayments travelers must make to settle up after a trip.
FR-002	Functional	The system shall allow a Group Traveler to record a shared expense by specifying the amount, currency, payer, and the participants who should split the cost.	High	Pass: A submitted expense with a payer and at least one participant appears in the trip ledger within 2 seconds and updates every participant's balance. Fail: An expense is accepted without a payer or without any participant.	Accurate, itemized expense data is a prerequisite for any correct settlement calculation.
FR-003	Functional	The system shall automatically convert every recorded expense from its original currency into the trip's base currency using the exchange rate in effect on the date the expense was logged.	High	Pass: A foreign-currency expense displays both its original amount and the converted base-currency amount, using the rate dated to its entry date. Fail: Two expenses logged on different dates in the same currency use an identical, stale rate when a newer rate existed.	Directly required by the problem statement (multi-currency support with daily exchange rates); without normalization to one currency, balances and settlement math are meaningless.
FR-004	Functional	The system shall allow a Trip Admin to create a trip, set its base currency, and add or remove Group Traveler members for the duration of the trip.	Medium	Pass: A Trip Admin creates a trip, adds at least two travelers, and removes a traveler with a zero balance. Fail: A traveler with a non-zero balance is removed without any warning or block.	Establishes the group and currency context that every other use case (expenses, conversion, settlement) depends on.
FR-005	Functional	The system shall allow any Group Traveler or Trip Admin to view, at any time, a real-time summary of each member's net balance (amount owed or owing) for the trip.	High	Pass: Reopening the settlement view after a new expense is added reflects the updated balances without any manual recalculation step. Fail: Balances shown are stale relative to the latest recorded expense.	Travelers need visibility into who owes what before the final settlement graph is generated, so data-entry mistakes can be caught early.
NFR-001	Nonfunctional	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Keeps the app responsive when several travelers request the settlement view at once, typically right at the end of a trip.
NFR-002	Nonfunctional	The system shall use a currency exchange rate that is no more than 24 hours old for every conversion; if the exchange-rate provider is unreachable, it shall fall back to the last successfully cached rate and flag affected expenses as rate-may-be-outdated.	Medium	Pass: When a simulated exchange-rate API call fails, the system completes the expense entry using the cached rate and shows the outdated-rate flag. Fail: Expense entry fails entirely, or a rate older than 24 hours is used silently with no flag.	Currency conversion is core to the app; an external API outage should degrade gracefully instead of blocking travelers from logging expenses.